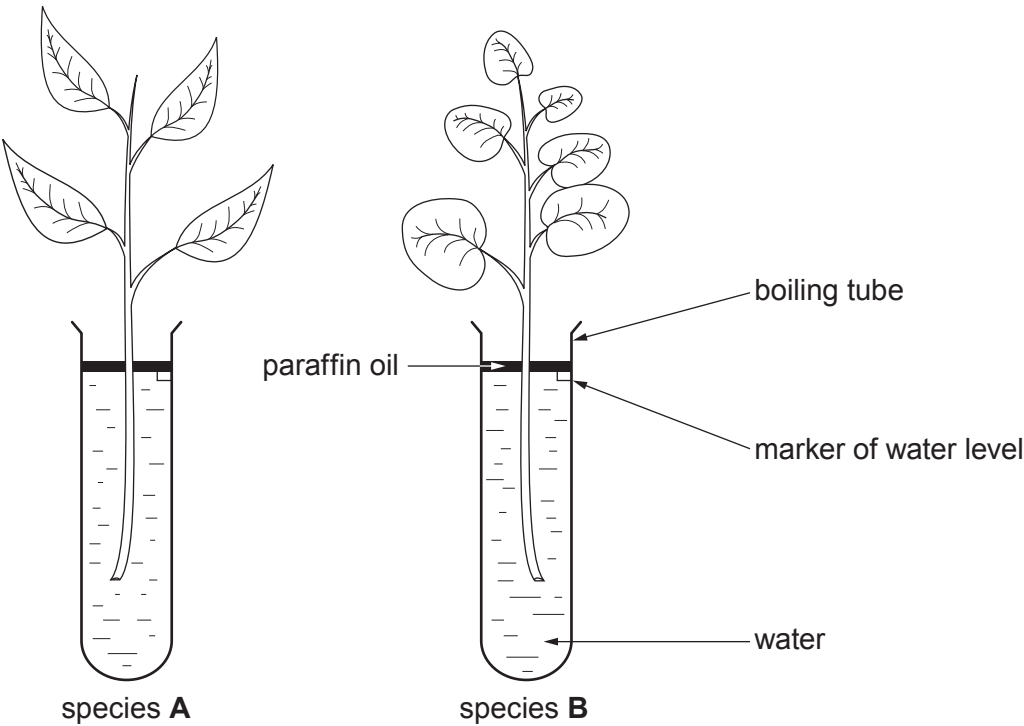
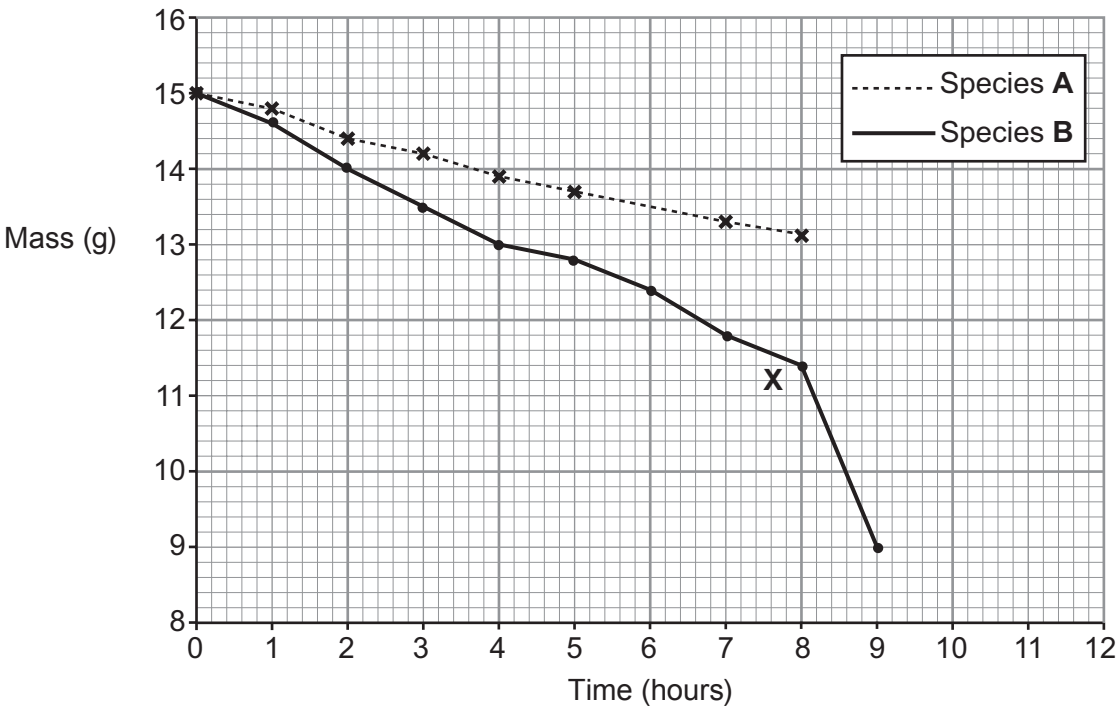


Answer all questions.

1. Shoots from two different species of plants, **A** and **B**, were placed in water in boiling tubes as shown below.



Both species were kept in the same conditions and their mass recorded at hourly intervals for eight hours. The results were recorded as line graphs shown below.



- (a) (i) Use the data to help you calculate the difference in the loss in mass between species **A** and **B** at 8 hours. [1]

Difference in loss of mass = g

- (ii) State the term given to the process responsible for this loss of mass in plants. [1]

- (iii) Suggest **two** reasons for the difference in loss of mass between the two different species, **A** and **B**. [2]

- (iv) State the purpose of the paraffin oil in the boiling tubes. [1]

- (b) Which **one** of the following treatments was applied to species **B**, at point **X** on the graph, to produce the result shown at 9 hours?

Underline the correct answer. [1]

- Enclosed in a bell jar
- Upper surface of leaves coated with Vaseline
- Half the leaves removed
- Air directed at the leaves by a fan

- (c) What would you expect to happen to the rate of loss of mass if species **A** and **B** had roots? Give a reason for your answer. [2]

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(d) State **one** feature of the digestive system which helps this part carry out each function named in (c). [2]

Function 1.

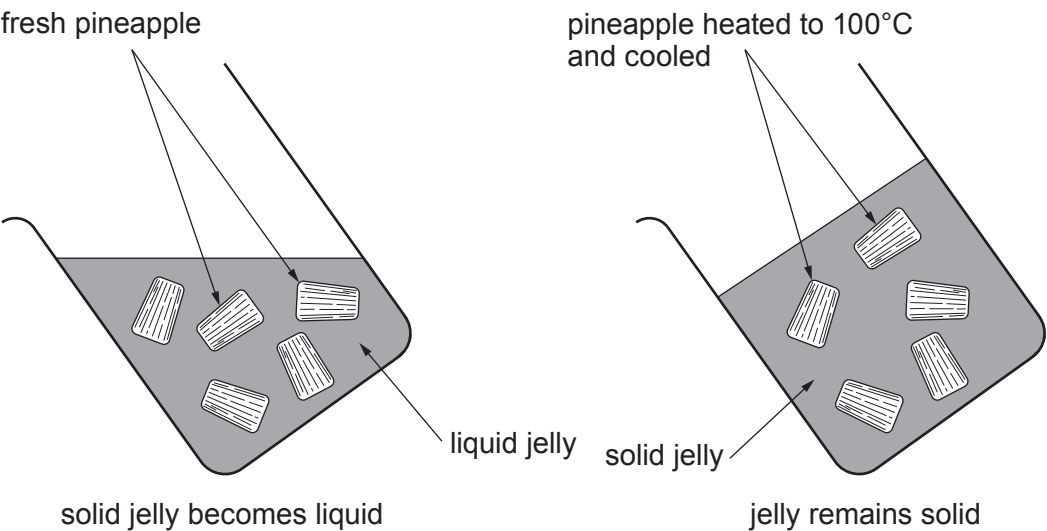
Function 2.

(e) Fresh pineapple is sometimes added to meat to make it tender. Fresh pineapple contains a protease enzyme.

Alan investigated how this took place. He used the following:

- fresh pineapple
- pineapple that had been heated to 100 °C and cooled
- a solid jelly made from protein called gelatin

Two samples of solid jelly were made and the pineapple was pushed into them. After 24 hours the results were shown below:



Explain the results with:

(i) fresh pineapple; [3]

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(ii) pineapple heated to 100°C and cooled. [3]

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